

Isolating individual spheroplasts for single-cell RNA extraction

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

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1. **Entire method performed between 0-4 °C** unless otherwise specified.
2. 10 mL mid-log culture transferred to 15 mL centrifuge tube.
3. Clusters pelleted by centrifugation at $4,000 \times g$ for 2 minutes. Discard supernatant.
4. Dissociate clusters by pipetting up and down ~50 times using a 100 μ L wide-bore pipette tip.
5. Gently crush clusters between two microscope slides: use the 100 μ L wide-bore pipette tip to drop 10 μ L of cluster pellet onto ~15 spots on a microscope slide, then gently place a second microscope slide on top of the first, then place a third slide on top of the second, perpendicular to it, forming a cross shape. Press lightly, then very gently remove the third and second slides.
6. Rinse crushed clusters from the first and second slides using 1 mL ice-cold Spheroplasting buffer (1 M sorbitol, 50 mM sodium phosphate pH 7.5, 10 mM EDTA, 2 mM DTT, 100 μ g/mL BSA) into a 2 mL tube.
7. Transfer 100 μ L of suspended crushed clusters to a 1.5 mL tube and pellet by centrifuging twice at $5,000 \times g$ for 30 seconds each time. Discard supernatant.
8. Resuspend pellet in 200 μ L of Spheroplasting buffer warmed to 30 °C and containing 15 U zymolyase.
9. Move the suspension to a tube suitable for end-to-end rotation (e.g., Sarstedt, Cat. No. 72.690.001).
10. Quickly flip this tube a few times to wet the inner walls, then incubate at 30 °C for exactly 20 minutes with constant end-to-end rotation at ~23 rpm to prevent clumping.
11. After incubation, immediately put the tube on ice.
12. Using a wide-bore pipette, move spheroplasts through a 10 μ m cell strainer into a new tube, and then through an 8 μ m cell strainer, to remove larger and smaller clumps respectively.
13. Pellet spheroplasts by spinning at $2,000 \times g$ for 5 minutes using a swinging-bucket centrifuge. Remove supernatant without disturbing the pellet.
14. Wash with ice-cold Wash Solution (1 M sorbitol, 10 mM Tris pH 8, 100 μ g/mL BSA) three times, using the same centrifuge settings as for pelleting the spheroplasts.
15. Resuspend cell pellet in 40 μ L Wash Solution.
16. Filter resuspension through an 8 μ m cell strainer into a new 1.5 mL tube, stored on ice.